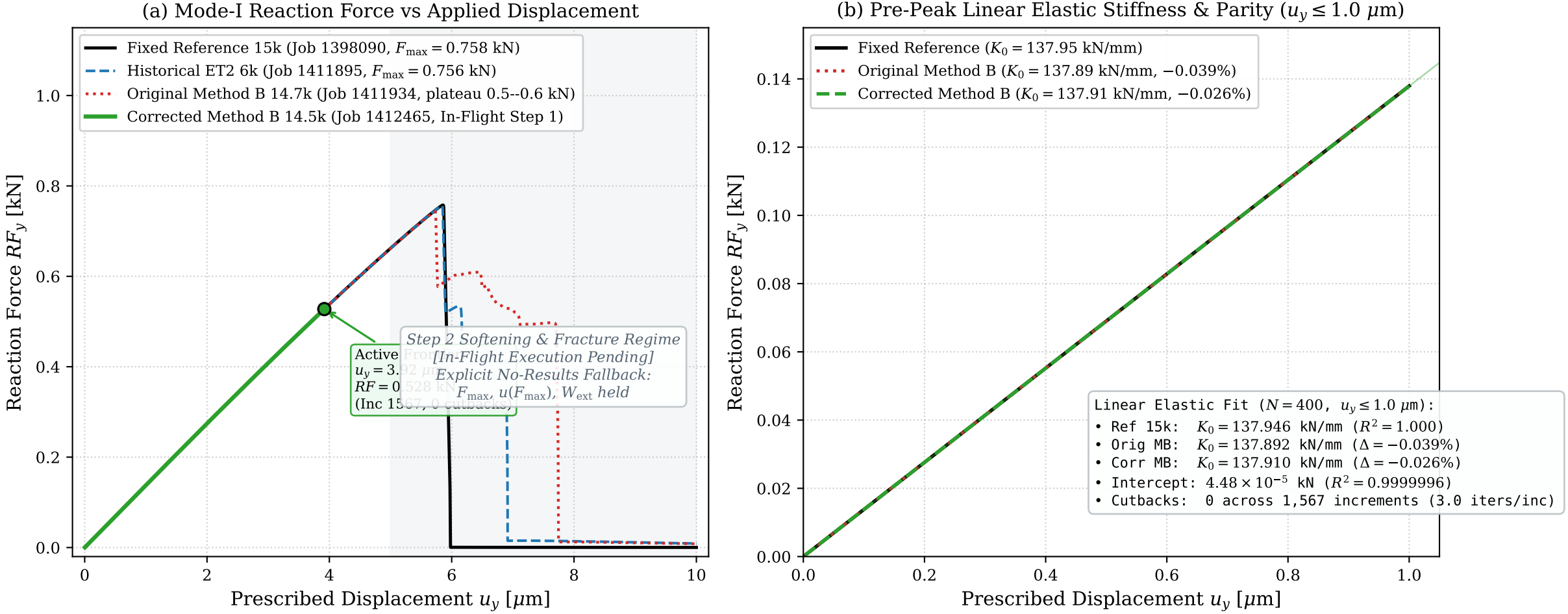

(a) Mode-I Reaction Force vs Applied Displacement
Fixed Reference 15k (Job 1398090, $F_{max} = 0.758$ kN)
Historical ET2 6k (Job 1411895, $F_{max} = 0.756$ kN)
Original Method B 14.7k (Job 1411934, plateau 0.5--0.6 kN)
Corrected Method B 14.5k (Job 1412465, In-Flight Step 1)
Step 2 Softening & Fracture Regime
[In-Flight Execution Pending]
Explicit No-Results Fallback:
F_{max}, $u(F_{max})$, W_{ext} held
Reaction Force RF_y [kN]
Prescribed Displacement u_y [μm]
(b) Pre-Peak Linear Elastic Stiffness & Parity ($u_y \leq 1.0$ μm)
Fixed Reference ($K_0 = 137.95$ kN/mm)
Original Method B ($K_0 = 137.89$ kN/mm, -0.039%)
Corrected Method B ($K_0 = 137.91$ kN/mm, -0.026%)
Linear Elastic Fit ($N = 400$, $u_y \leq 1.0$ μm):
• Ref 15k: $K_0 = 137.946$ kN/mm ($R^2 = 1.000$)
• Orig MB: $K_0 = 137.892$ kN/mm ($\Delta = -0.039\%$)
• Corr MB: $K_0 = 137.910$ kN/mm ($\Delta = -0.026\%$)
• Intercept: 4.48×10^{-5} kN ($R^2 = 0.9999996$)
• Cutbacks: 0 across 1,567 increments (3.0 iters/inc)
Reaction Force RF_y [kN]
Prescribed Displacement u_y [μm]